Not All Chatbots Teach: Evidence for Pedagogical Design in AI-Assisted Technical Education

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Abstract

As generative AI tools like ChatGPT become embedded in technical education, a critical challenge emerges: how can we ensure these tools foster learning rather than bypass it? This study provides empirical evidence that pedagogical design, not merely model access, determines the educational value of AI assistants. We developed a freely available custom conversational AI tool that embeds metacognitive scaffolding through structured prompts grounded in the Feynman Technique and learning science literature. In a quasi-experimental study within an undergraduate data structures course ($N = 36$), students using this structured AI assistant significantly outperformed peers using the same interface configured as a minimally prompted ChatGPT wrapper (92.7 vs. 74.3, $p < .001$). Gains were especially strong in abstraction, technical justification, and documentation, which are skills critical across software engineering, IT, and cybersecurity. These findings underscore a key insight: AI-integrated learning environments must be intentionally designed to prompt reflection, prediction, and explanation. By aligning AI interactions with evidence-based pedagogy, our framework demonstrates how to develop conceptual understanding, reduce automation bias, and support equitable learning outcomes as AI reshapes computing education.

CCS Concepts

• **Applied computing** → **Education**; *Collaborative learning*; • **Human-centered computing** → *Collaborative interaction*; • **Computing methodologies** → Natural language processing.

Keywords

Pedagogical AI Design, Metacognitive Scaffolding, Technical Education, Feynman Technique, AI-Assisted Learning

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1 Introduction

Large language model (LLM) tools such as *ChatGPT* have quickly evolved from experimental novelty to default companion in technical coursework. A recent SIGCSE survey reports that 79% of computer science undergraduates regularly use generative AI for programming tasks, with adoption rates surging by senior year [8]. Yet while access is ubiquitous, educational outcomes remain uneven. Studies have linked unstructured LLM use to trial-and-error troubleshooting [26], lower self-efficacy [9], and reduced conceptual transfer [17]. In contrast, structured AI tutors that embed learning-science-informed prompts have shown promise: boosting student reflection [16], metacognition [23], and decision-making under uncertainty [24].

Despite growing interest in AI-driven pedagogy, empirical comparisons between structured and unstructured AI support within the same instructional context remain rare. As institutions race to embed LLMs into education, from programming to network design and cybersecurity, the core pedagogical challenge persists: How can we ensure that AI tools facilitate deeper learning, not shortcut it?

Our study addresses this challenge through a quasi-experimental design embedded in an undergraduate Data Structures course ($N = 36$). We developed an AI-assisted learning tool with configurable pedagogical scaffolding, allowing features such as Feynman-style prompts and conceptual checkpoints to be activated or deactivated. In our study, two student sections completed identical assignments under the same instructor and grading criteria, but with one critical difference: the treatment group used the tool with scaffolding enabled, whereas the control group used the same interface configured solely as a wrapper for a minimally prompted ChatGPT instance. This design isolates the impact of pedagogical structure from LLM capability.

Findings were dramatic. Students using the structured assistant scored an average of 92.7 on the culminating project, significantly outperforming peers in the unstructured condition (mean = 74.3,

$p < .001$). Gains were particularly strong in areas requiring abstraction, justification, and documentation, skills foundational across CS, IT, and cybersecurity domains. Moreover, performance variability decreased in the treatment group, suggesting promising equity benefits.

These results offer two essential insights. First, the educational value of generative AI depends not only on model power, but on intentional integration with pedagogical best practices. Second, metacognitive scaffolding, rooted in well-established theories like the Feynman Technique and self-regulated learning can reliably reshape how students engage with AI tools.

Although our intervention was validated in a data structures project focused on selecting and justifying abstractions, its design principles are broadly transferable. Whether learners are evaluating protocol trade-offs in IT labs or assessing security configurations in adversarial contexts, the need to explain, predict, and justify remains central. As LLMs become part of instructional infrastructure, this work offers a theory-grounded, evidence-based blueprint for AI integration.

The remainder of this paper is organized as follows. Section 2 reviews prior work on AI in technical education and pedagogical scaffolding. Section 3 describes our scaffolding tool, and Section 4 details the study design and implementation context. Section 5 reports key performance findings, Section 6 addresses validity threats, and Section 7 offers conclusions and directions for future research in technical education with AI.

2 Related Work

The rapid adoption of large language models (LLMs) like ChatGPT has generated both enthusiasm and concern across technical education domains. While many students rely on these tools for code generation, debugging, and explanation, recent research suggests their educational impact depends heavily on the structure of interaction [9, 17].

2.1 Limitations of Unstructured AI Use

Unstructured AI interactions often lead to surface-level engagement. Without prompting for reasoning or self-explanation, learners may bypass core problem-solving processes [13, 26]. Instructors report declining peer collaboration and inconsistent outcomes when LLM tools are used without pedagogical guidance [9, 17].

These concerns are particularly acute in cybersecurity education. Students using generic AI assistance for security configuration have implemented flawed systems while believing them secure, highlighting risks of automation bias [11, 22]. Similarly, in networked systems and IT administration, students often struggle to diagnose or justify configurations produced by AI tools, especially when reasoning steps are omitted [12, 25].

2.2 Structured Pedagogical Design

In contrast, structured AI tools designed with pedagogical scaffolds have consistently shown greater effectiveness. By prompting for explanation, justification, or prediction, structured assistants support metacognitive engagement and critical thinking [3, 5]. In STEM and IT domains, such scaffolding helps students reason about trade-offs, systems behavior, and design decisions rather than copying AI-generated outputs [7, 18].

Notably, implementations like SRLbot [19] and structured models grounded in scientific reasoning frameworks (e.g., Predict, Observe, Explain) have improved inquiry skills and conceptual understanding in controlled studies [6].

2.3 Learning Science Foundations

These effects are supported by foundational learning theory. Techniques such as the Feynman Method and self-regulated learning emphasize articulation, prediction, and revision, skills embedded in prompt-driven scaffolds [2, 23]. Human-centric explainable AI (XAI) systems have also shown promise for technical education by supporting iterative, transparent learning aligned with mental model development [10].

2.4 Implications for Technical Education

Across programming, IT, and cybersecurity, the evidence converges: unstructured AI use can hinder conceptual growth, while structured support promotes deeper understanding. As technical education increasingly includes LLM-mediated experiences, prompt design and pedagogical scaffolding emerge as central factors for ensuring equitable, meaningful learning.

3 Pedagogical Scaffolding Tool

Effective AI tool design requires more than model access, it demands intentional structuring of learner interaction. Without pedagogical scaffolding, LLM-based assistants can reinforce automation bias, shortcut reasoning, and promote shallow learning [11, 13]. To address this need, we developed an online tool that employs a modular prompt framework informed by learning science principles and structured to elicit explanation, prediction, and reflection. The tool is freely accessible at [anonymized URL].

3.1 Prompt Sequence Framework

Our assistant uses a four-phase conversational scaffold:

(1) **Explain** – Students describe a concept or intent using their own language.
(2) **Predict** – They forecast the behavior or outcome of a decision.
(3) **Reflect** – After executing or observing, they revisit their belief and compare it to what occurred.
(4) **Revise** – Students update their reasoning or approach based on structured questioning.

This cycle operationalizes the Feynman Technique and self-regulated learning principles [2, 20] to guide learners toward conceptual clarity.

3.2 Prompt Examples Beyond Programming

While tested in a programming context, the framework generalizes across technical disciplines. Table 1 shows a representative scaffold from cybersecurity.

Table 1: Prompt types across reasoning phases (cybersecurity example)

Phase	Example Prompt
Explain	"Describe symmetric vs. asymmetric encryption. Where would you use each?"
Predict	"What risks arise if Company X switches from HTTPS to HTTP?"
Reflect	"Now that you captured traffic with Wireshark, did the results match your expectations?"
Revise	"Given your findings, would you adjust the firewall setup? Why or why not?"

3.3 Prompt Categories and Flow

Each prompt serves one of four pedagogical intents:

- **Concept Articulation**: Reframe known content in the learner's own words.
- **System Reasoning**: Apply knowledge under constraints or conditions.
- **Diagnostic**: Identify gaps or surface faulty assumptions.
- **Justification**: Construct a rationale grounded in principles or trade-offs.

Conversations are lightly stateful: earlier answers shape follow-up scaffolds (e.g., "Earlier you said port 80 was encrypted. Does your Wireshark log confirm that?"). Students must complete reflection prompts before accessing solution suggestions, nudging deeper engagement and discouraging premature answer-seeking [9, 17].

3.4 Transferable Scaffolding for Technical Domains

The Explain–Predict–Reflect–Revise structure is applicable beyond programming. In networking, it can help students reason through a misconfigured subnet; in IT, predict the effects of a DNS error. These technical reasoning loops form the basis for judgment in a range of domains [18, 25].

3.5 Design Summary

The core insight is architectural: prompt design defines educational value. Our assistant ensures that every interaction demands cognitive effort, turning reactive querying into active reasoning. This design aligns with ACM IT2017 [4], CS2023 [14], and modern learning science goals for equitable, reflective technical education [23].

4 Study Design

We conducted a quasi-experimental study to isolate the educational impact of structured AI scaffolding vs. unstructured chatbot support in technical problem-solving.

4.1 Participants and Setting

Participants ($N = 36$) were undergraduates in two sections of a data structures course at a U.S. liberal arts university in the Northwestern United States during the Spring 2025 semester. One section ($n = 19$) used a structured, pedagogically-designed AI assistant; the other ($n = 17$) accessed standard ChatGPT. Both groups had the same instructor, assignments, and prerequisite backgrounds.

4.2 Assignment and Context

The intervention focused on a capstone assignment, the *Bistro Ordering System*, requiring:

- Selection/justification of data structures for system modules,
- Implementation of a working text-based project,
- Documentation of design decisions.

This open-ended context, requiring synthesis and abstraction, was chosen to test the potential of metacognitive scaffolding.

4.3 Treatment Conditions

Structured AI Assistant: Students completed a "Project Design Journey" of sequential prompts (explain, predict, reflect, revise), inspired by the Feynman Technique. Advancement required responding thoughtfully at each scaffolded stage; system state-tracking ensured consistent exposure and engagement.

Unstructured ChatGPT: Control students accessed the same user interface as the treatment group; however, it served solely as a wrapper for a minimally prompted instance of ChatGPT. The interface provided only essential contextual information related to the assignment and permitted unrestricted AI queries. No scaffolding mechanisms or reflective prompts were incorporated.

4.4 Procedure and Measures

- Both sections received the assignment simultaneously and used their assigned tool, with extra credit encouraging engagement.
- Projects were collected after three weeks and graded by blinded TAs using a standardized rubric.
- Metrics included overall and subcategory project scores, tool usage logs, and brief post-project surveys.

4.5 Variables

Independent Variable: AI condition (structured vs. unstructured).

Dependent Variables: Rubric-based scores (functional implementation, data structure use, code quality, report accuracy, documentation; 20 points each).

Controls: Instructor, content, grading, and incentives standardized across groups.

4.6 Ethics and Limitations

The study was IRB-approved, with voluntary participation and de-identified analysis. Section assignment (not randomized) and non-uniform tool engagement in the control group are recognized as internal validity threats, addressed further in Section 6.

Summary: This design allowed for a rigorous yet pragmatic test of whether structured scaffolding, apart from model sophistication, improves conceptual performance in technical coursework.

5 Results and Discussion

We report on the impact of pedagogical scaffolding in AI tool design by comparing project performance between students who used the structured, metacognitive assistant and those who used

a generic ChatGPT interface. We analyze both overall scores and rubric-specific dimensions to identify where structured interactions generated the largest benefits for learning.

5.1 Overall Performance Gains

Students using the structured AI assistant significantly outperformed their peers. The average project score in the structured condition was 92.7 (SD = 3.8), compared to 74.3 (SD = 10.2) in the unstructured condition. An independent samples t-test confirmed this difference is highly significant ($t = 6.93$, $df \approx 22.7$, $p < .000001$), with a large effect size (Cohen's $d = 2.14$).

Equally notable was the lower variance in the treatment group. Students supported by the structured assistant produced more uniformly high-quality work, suggesting that pedagogical scaffolding may not only raise performance on average, but also support more equitable outcomes across diverse learners.

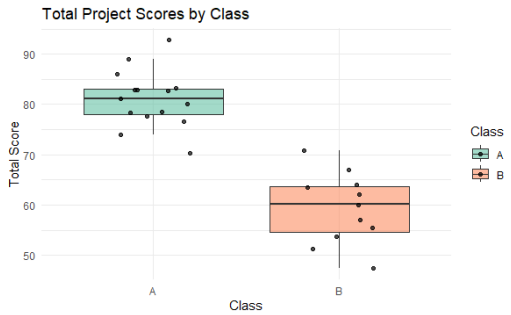

Figure 1: Boxplot of total project scores by condition. Structured tool users show higher average and lower variance.

5.2 Category-Level Performance

Analyzing rubric components reveals where the pedagogical design had the strongest effect. Across five graded dimensions, students using the structured tool significantly outperformed their peers in four categories:

- **Functional Implementation** – Programs were more complete and executed intended features correctly ($p < .0001$).
- **Data Structure Usage** – Students selected more appropriate and efficient data structures, justifying their choices more effectively ($p = .0002$).
- **Report Accuracy** – Explanations of program logic were clearer, more precise, and closely aligned to actual behavior ($p = .0005$).
- **Documentation Quality** – README files and code comments reflected thoughtful design rationales and deeper conceptual understanding ($p < .0001$).

There was no statistically significant difference in **Code Modularity and Style** ($p = .27$), suggesting that formatting conventions and structural organization were adopted similarly across groups.

5.3 Interpretation and Pedagogical Implications

These outcomes offer strong support for our central hypothesis: that learning benefits from AI tools depend not on access alone,

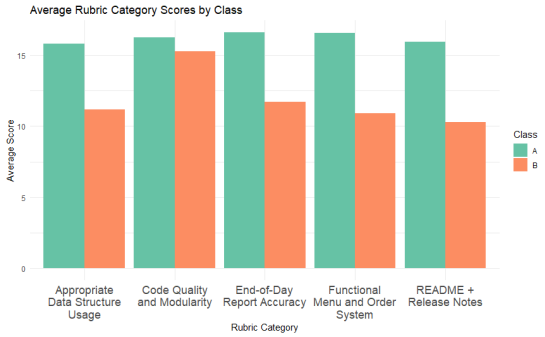

Figure 2: Average rubric category scores by condition. Structured tool users scored higher in all dimensions except code modularity.

but on pedagogical design. The most substantial gains occurred in areas involving abstraction, justification, and documentation, skills foundational to technical education but often undermined by unstructured AI use [9, 15, 24].

The large effect size reflects the alignment between our scaffolding approach and learners' needs during the assignment. Rather than offering answers, the structured assistant guided students to reason through problem constraints, predict outcomes, and justify design choices, processes established as cornerstones of conceptual learning [16, 23]. This metacognitive engagement appeared critical in helping students avoid AI-induced automation bias [11], reinforcing deep learning over superficial solution-seeking.

Our tool operationalized the Feynman Technique in real time, requiring students to externalize their reasoning as a condition of tool use. This approach mirrors findings by Ambion et al. [2] and Akinola et al. [1], who found that explanation-based learning leads to better retention and problem transfer in technical disciplines.

The observed reduction in performance variance also implicates equity benefits. Structured scaffolding may help bridge achievement gaps by supporting students who might otherwise struggle with open-ended AI use [21]. Unlike generic chat interfaces, which reward existing prior knowledge and strategic querying—our assistant ensures that all students engage in reflective practice regardless of expertise.

Figure 3 provides an illustrative example. Unlike unstructured tools that supplied implementation code directly, the scaffolded assistant responded with a series of guided prompts requiring students to articulate system behavior expectations before receiving support. In this interaction, pedagogical structure becomes not a constraint, but a catalyst for disciplinary thinking.

In sum, these findings validate our assertion that LLM-powered tools can meaningfully contribute to technical education, but only when intentionally designed to engage the learner's reasoning processes. Our structured tool helped transform a potentially passive coding helper into an interactive pedagogical partner.

6 Threats to Validity

While our findings support the value of pedagogically structured AI tools, several threats to validity warrant consideration.

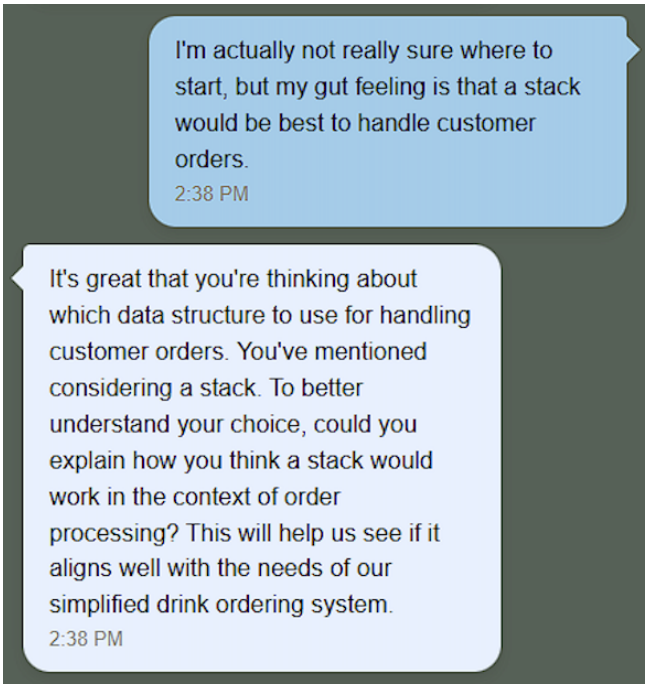

Figure 3: Interaction snapshot from the structured assistant using the Feynman Technique. Prompts guide students to predict outcomes and reason before implementation.

6.1 Internal Validity

As a quasi-experimental study, participants were assigned by section rather than randomly, raising potential selection bias. Although both groups were matched on instructor, content, and prior experience, uncontrolled differences (e.g., scheduling, cohort dynamics) may have influenced outcomes.

Tool usage requirements also varied: structured tool users had to complete all scaffolding steps (for extra credit), ensuring deeper engagement, while control students used a minimally pre-prompted ChatGPT wrapper variably. This introduces a usage–engagement confound despite our aim to reflect authentic classroom conditions. Although projects were graded blindly by teaching assistants using a shared rubric, instructor awareness of group assignment adds a minor bias risk.

6.2 Construct Validity

We measured learning impact via rubric-based project scores, which capture product quality but not metacognitive growth or long-term transfer. Though our measures align with practical course outcomes, they underrepresent internal strategy development.

Our custom tool also combines multiple pedagogical strategies, including Feynman-based explanation, prediction cycles, and domain-aware scaffolds, limiting our ability to isolate specific mechanisms. While we suspect these components interact synergistically, controlled ablation studies would better establish causal drivers.

6.3 External Validity

Our sample (N=36) came from a single liberal arts institution and one course section per condition. While we selected a system-design task broadly relevant to CS, IT, and cybersecurity, the specific context limits generalizability. Replication across institutions and disciplines is needed.

Model versions were synchronized during the study, but LLM behavior evolves rapidly. Future replications may encounter different outputs or user expectations, affecting both tool effects and student interpretation.

6.4 Statistical Conclusion Validity

The large performance gap between groups (Cohen's $d = 2.14$) invites further scrutiny. Low variance in the treatment group may conceal differences in who benefited most; high variance in the control group may reflect uneven tool use or varied comprehension. Sample size constraints also limit finer-grained statistical inference.

In summary, these validity threats highlight areas for cautious interpretation and future exploration. However, they do not undermine our key claim: that structured scaffolding significantly enhances the educational value of AI tools. The alignment between performance outcomes and learning science principles provides converging evidence for pedagogical design as a critical success factor in LLM-supported technical education.

7 Conclusion and Future Work

Digital tools like ChatGPT are reshaping how students engage with technical material, but their impact is rarely pedagogically neutral. In practice, unstructured AI use often replaces rather than supports the learning process. This study provides empirical evidence that pedagogically structured AI tools can shift that dynamic—from passive automation to active learning.

7.1 Key Contributions

This work makes three contributions to technical education:

1. Empirical Validation of Pedagogical Design: In a quasi-experimental study within a real-world Data Structures course, we demonstrate that AI tools embedded with metacognitive scaffolding significantly improve student performance, particularly in abstraction, justification, and documentation. The observed effect size (Cohen's $d = 2.14$) reflects both statistical robustness and substantial pedagogical impact.

2. Transferable Design Framework: We implement an explain–predict–reflect–revise workflow grounded in the Feynman Technique through structured dialog and prompt sequencing. This scaffolding model is adaptable to other technical domains where conceptual decision-making is central, such as cybersecurity, systems design, and network configuration.

3. Practical Blueprint for AI Tool Development: We show how pedagogical principles can be systematically encoded into an accessible, conversational interface, aligning with ACM IT2017 and CS2023 guidelines for promoting metacognition and equitable engagement. Our tool, freely available for educational and research use, serves not just as a chatbot, but as a replicable blueprint for curriculum-aligned AI integration.

7.2 Directions for Future Work

This early-stage study opens multiple pathways for advancing AI-integrated education:

- **Randomized and Cross-Institutional Studies:** Expand to diverse institutions and student populations to test generalizability and causal strength.
- **Domain-Specific Scaffolding:** Tailor the structured framework to disciplines such as cybersecurity, IT systems deployment, and networking, while retaining metacognitive elements.
- **Longitudinal Learning Impact:** Measure long-term retention, transfer to novel contexts, and professional application beyond immediate course performance.
- **Adaptive Pedagogical Agents:** Personalize scaffolding by fading prompts as expertise grows or increasing support when confusion is detected.
- **Tool Authoring Environments for Instructors:** Build lightweight platforms enabling faculty to design structured AI assistants from templates, checklists, and assessment integrations.

7.3 Final Reflection

As generative AI becomes embedded in computing education, its impact will hinge less on whether it is used and more on how it is designed. When grounded in learning science, AI can shift from shortcut to catalyst, prompting learners to explain, predict, and refine their thinking. This study delivers a validated, theory-driven blueprint—freely available for education and research, for integrating AI in ways that elevate learning outcomes and advance equity in the evolving craft of technical education.

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